Supplier master data: quality report

Before and after cleaning · SECOP II registered suppliers in Boyacá, Colombia, cross-checked with the business registry (RUES) · data extracted on 2026-10-26

4,164	3,569	103	492	1,142
records in the raw file	organisations in the clean file	duplicate records removed	excluded for privacy	records with a problem flagged

Before vs after: share of records with a usable value

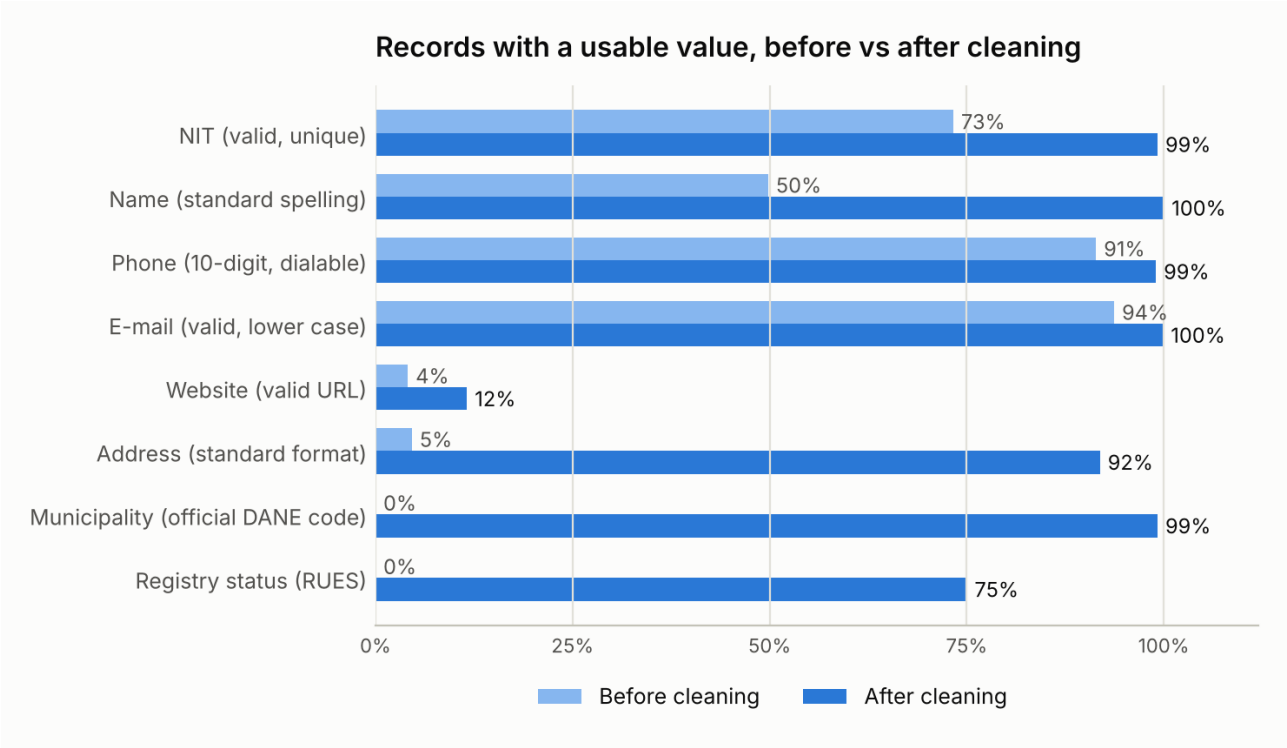
Field	Before	After	What changed
NIT (valid, unique)	73.3%	**99.2%**	Check digit split off and verified; duplicates removed
Name (standard spelling)	49.8%	**100.0%**	Upper case, one spelling per legal form
Phone (10-digit, dialable)	91.4%	**99.0%**	Old 7-digit landlines converted to 608 + number
E-mail (valid, lower case)	93.7%	**99.9%**	Lower case, typos fixed, invalid blanked
Website (valid URL)	4.1%	**11.6%**	https:// added, typos fixed (most suppliers have none)
Address (standard format)	4.6%	**91.9%**	CALLE/CRA/No. -> CL/KR/#; rural addresses accepted
Municipality (official DANE code)	0.0%	**99.2%**	Matched to the DANE list (DIVIPOLA)
Registry status (RUES)	0.0%	**75.0%**	Status, registration date and renewal added

Before = the 4,164 raw records, used as they come. After = the 3,569 records in the clean file.

Main rules applied (full log in the Excel file)

Rule	Affected	Detail
Trim spaces and blank out placeholders	18,345 cells	18,062 placeholder cells ('No Provisto', 'No Definido'...) set to empty; 283 cells had extra spaces
Split NIT and check digit	614 rows	10-digit values were NIT + check digit; check digit verified with the DIAN modulo-11 rule
Exclude personal ID numbers	428 rows	Records whose ID is a personal ID (cédula), not an organisation NIT: excluded so no personal data is published
Privacy: names that look like a person's name	93 rows	29 replaced with the legal name from RUES; 64 without a registry record excluded (heuristic: 2-4 plain words, no acronym and no word that organisations use)
Remove duplicate NITs	103 rows	103 NITs appeared more than once; kept the record with the most contact fields, then the most recent
Check legal form vs company type	157 rows (corrected)	SECOP company type disagrees with the legal form in RUES (or in the name suffix when RUES has no record); legal_form column holds the corrected value
Standardise phone numbers	3,532 rows	To +57 format; old 7-digit landlines get the 608 area prefix (2021 numbering plan); fax used when the phone was missing (4)
Privacy: mask personal-looking e-mails	821 rows	Mailbox is not a role (info@, ventas@...) and does not contain the organisation's name, so it may belong to a person
Standardise addresses	3,023 rows	Colombian abbreviations (CALLE/CLL -> CL, CARRERA/CRA -> KR, No./N° -> #)
Cross-check with the business registry (RUES)	2,675 rows	2,675 found; status, registration date and last renewal added; 469 active records not renewed for 2026

Quality before and after cleaning



Registry cross-check (RUES)

Registry status	Organisations
Active	2,506
Not found in RUES	894
Active (moved from another chamber)	139
Cancelled	28
Moved to another chamber	2

Most frequent problems flagged	Records
registration not renewed for 2026	469
no street type or number	284
trade name differs from the legal name in RUES	251
SECOP type 'Sociedad Anónima Abierta colombiana' but the registry says SAS	80
company is in liquidation (RUES name)	55
municipality missing	30
SECOP type 'Sociedad Anónima Cerrada Colombiana' but the registry says SAS	28
invalid length or format	28

For a procurement team: 469 suppliers with an active registration did not renew it for 2026, 55 companies are in liquidation and 28 registrations are cancelled. These are the first records to check before signing a contract.

Method and privacy

Python (pandas) pipeline with unit-tested rules: NIT check digit (DIAN modulo 11), Colombian phone plan (2021), e-mail and URL syntax, standard address abbreviations, DANE municipality codes, dates read from three different formats, and a registry cross-check by NIT. Values that cannot be fixed with certainty are flagged, never guessed. Only organisations are published: personal IDs and person-like names are excluded, 821 personal-looking e-mails are masked and mobile numbers are partly hidden.

Limitations

The person-name check is a heuristic and may also remove a few organisations named after people. Organisations not found in RUES (community action boards, consortia, public bodies) are often registered elsewhere, so 'not found' is not an error by itself. Phones and websites were not called or visited.

Sources

SECOP II - Proveedores Registrados (Colombia Compra Eficiente) and RUES (Confecámaras), datos.gov.co, CC BY-SA 4.0. DIVIPOLA (DANE), datos.gov.co, CC BY-SA 4.0.